

FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10: 2.1 PROPERTIES OF WAVES – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How do sound waves reflect, and why do some places produce stronger echoes than others? Use what you learned in Lessons 1 to 12 to write your Final Explanation. Answer each section in the space under it. Use the Echo Table and the Surface Table below, and show every calculation with units. Speed of sound in air is 340 m/s. Use the echo-distance formula $d = vt/2$, where t is the total time from the sound leaving to the sound returning and d is the one-way distance. Write in full sentences and use Claim, Evidence and Reasoning where asked.

ECHO TABLE

Site	Place	Distance to nearest large hard surface	What the learners heard
A	Hell's Gate cliff face	85 m	Clear echo
B	Ngong Hills valley wall	170 m	Clear echo
C	Kisumu school hall	12 m	Loud, muddy, drawn-out sound
D	Nairobi school open grounds	300 m (classroom block)	No echo

SURFACE TABLE

Surface	Approximate share of sound energy reflected
Cliff rock	90%
Painted concrete wall	85%
Wooden board	60%
Curtains or cloth	15%

Thick grass or foam mattress	10%
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Part 1: What is happening to the sound? (Claim)

Wanjiru shouts towards the cliff at Hell's Gate and hears her voice come back. Describe what the sound wave does from the moment she shouts until she hears the echo. In your answer: (a) say what kind of wave sound is and what it needs to travel; (b) describe the path of the sound, using the idea of a ray; (c) state the two laws of reflection and say how they apply to the sound at the cliff; (d) say what we mean by an echo. Draw a labelled ray diagram showing the normal, the incident ray, the reflected ray, i and r .

Write your answer here:

Part 2: Delay and distance (Evidence)

Use the echo-distance formula to work out the following. (a) Show that the delay for the echo at Hell's Gate (site A) is 0.50 s. (b) Calculate the delay for Ngong Hills (site B). (c) Calculate the delay for the Kisumu hall (site C) and the one for the open Nairobi grounds (site D). (d) Explain why the ear needs a delay of at least 0.1 s to hear a distinct echo, and calculate the smallest distance to a reflector that gives a distinct echo. (e) Kiprono stands in front of a long school wall and times 10 clap-echo intervals as 2.94 s. Find the distance to the wall.

Write your answer here:

Part 3: Strong echoes and weak echoes (Reasoning)

Delay and distance alone do not explain every site. Use the Echo Table and the Surface Table to answer: (a) Why do Hell's Gate and Ngong Hills give clear, strong echoes? (b) Why is the sound in the Kisumu hall loud, muddy and drawn out? Use the words reverberation and interference in your answer. (c) Site D has a delay of 1.76 s, which is longer than 0.1 s, yet there is no echo. Explain why, using the ideas of spreading (diffraction) and absorption. (d) Which surface in the Surface Table would give the strongest echo, and which the weakest? Give a reason.

Write your answer here:

Part 4: Other wave behaviours at a boundary

When a wave meets a boundary, different things can happen. (a) Explain the difference between reflection and refraction, and say which of them returns the sound to the learners at Hell's Gate. (b) Light going from air into water at $i = 45$ degrees bends to $r = 32$ degrees. State what this tells you about the speed of light in water and why the light bends. (c) Sound can turn the corner of a 1.0 m doorway but a torch beam makes a sharp patch. Use the wavelength of a 340 Hz sound and the wavelength of light (5.5×10^{-7} m) to explain this.

Write your answer here:

Part 5: Final Explanation and applying the idea

Write your Final Explanation of the driving question: How do sound waves reflect, and why do some places produce stronger echoes than others? Use Claim, Evidence and Reasoning. Your answer must cover all four sites (A to D) and use at least three numbers from the tables. Then answer: (a) Mr. Mwangi asks you to reduce the reverberation in the Kisumu hall. Suggest a treatment from the Surface Table and explain how it works. (b) Give one way the echo idea is used in medicine and calculate the time for an ultrasound pulse (1540 m/s) to return from an organ boundary 7.7 cm deep.

Write your answer here:

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim: describing reflection and the laws of reflection	Clearly states that sound is a mechanical wave that travels as a ray, reflects at the surface and returns; states both laws of reflection correctly with angles measured from the normal; the ray diagram is fully labelled and accurate.	States that sound reflects and returns and gives at least one law of reflection correctly; the diagram has most labels but may miss the normal or one angle.	Says sound bounces back but does not state the laws, or measures angles from the surface; the diagram is missing or unclear.
Evidence: calculations with $d = vt/2$ and the 17 m condition	All delays (0.50 s, 1.00 s, 0.07 s, 1.76 s), the 17 m minimum distance and the 50 m field distance are correct, with working and units shown.	Most calculations are correct with the right formula, but there are small slips in arithmetic or missing units.	Uses the formula incorrectly (for example forgetting to halve or double) or gives several wrong values.
Reasoning: explaining the four sites using surfaces, reverberation, spreading and	Explains all four sites correctly, linking distance, delay, surface reflection percentages, reverberation and	Explains most sites correctly but leaves out one idea, such as absorption or the overlapping reflections.	Gives only descriptions of what was heard, or explains with ideas not supported by the data.

absorption	interference in the hall, and spreading and absorption on the grounds.		
Connecting other wave behaviours and applications	Correctly separates reflection from refraction, explains bending using speed change, uses wavelengths to explain diffraction, and calculates the ultrasound echo time of 100 microseconds.	Gets most of the distinctions right and does most of the wavelength or ultrasound working correctly.	Confuses reflection with refraction or does not use wavelength to explain diffraction.
Communication using Claim, Evidence and Reasoning	The Final Explanation has a clear claim, uses at least three numbers from the tables as evidence, covers all four sites, and uses scientific terms correctly and consistently.	Has a claim and some evidence, covers three or four sites, and mostly uses correct terms.	Claim, evidence and reasoning are mixed together or missing, and few data are used.

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